Quantum Mechanics Homework 6 Solutions

Lectures 13–14

Problem 1. Oscillator ladder and first excited state [20 points]

(a) Use

$$\hat{H}\hat{a}^\dagger = [\hat{H}, \hat{a}^\dagger] + \hat{a}^\dagger \hat{H}. \tag{1}$$

Acting on ϕ_n gives

$$\hat{H}(\hat{a}^\dagger \phi_n) = \left([\hat{H}, \hat{a}^\dagger] + \hat{a}^\dagger \hat{H}\right)\phi_n \tag{2}$$

$$= \hbar\omega \hat{a}^\dagger \phi_n + E_n \hat{a}^\dagger \phi_n \tag{3}$$

$$= (E_n + \hbar\omega)(\hat{a}^\dagger \phi_n). \tag{4}$$

Therefore, provided it is nonzero,

$$\boxed{\hat{a}^\dagger \phi_n \text{ is an energy eigenfunction with energy } E_n + \hbar\omega.} \tag{5}$$

(b) For a normalized state ϕ,

$$\int \phi^* \hat{H} \phi \, dx = \hbar\omega \int \phi^* \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right)\phi \, dx \tag{6}$$

$$= \hbar\omega \left[\int \phi^* \hat{a}^\dagger (\hat{a}\phi)\, dx + \frac{1}{2}\int |\phi|^2 \, dx\right] \tag{7}$$

$$= \hbar\omega \left[\int (\hat{a}\phi)^* (\hat{a}\phi)\, dx + \frac{1}{2}\right] \tag{8}$$

$$= \boxed{\hbar\omega \left[\int |\hat{a}\phi|^2 \, dx + \frac{1}{2}\right]}. \tag{9}$$

Since $\int |\hat{a}\phi|^2 dx \geq 0$,

$$\int \phi^* \hat{H} \phi \, dx \geq \frac{1}{2}\hbar\omega. \tag{10}$$

For the ground state, $\hat{a}\phi_0 = 0$, so the nonnegative term vanishes. Hence

$$\boxed{E_0 = \frac{1}{2}\hbar\omega.} \tag{11}$$

(c) Differentiate the ground-state Gaussian:

$$\frac{d\phi_0}{dx} = -\frac{2x}{x_0^2}\phi_0. \tag{12}$$

Since $x_0 p_0 = 2\hbar$,

$$\frac{\hbar}{p_0} = \frac{x_0}{2}. \tag{13}$$

Therefore

$$\hat{a}^\dagger \phi_0 = \left(\frac{x}{x_0} - \frac{\hbar}{p_0} \frac{d}{dx} \right) \phi_0 \tag{14}$$

$$= \left(\frac{x}{x_0} + \frac{x_0}{2} \frac{2x}{x_0^2} \right) \phi_0 \tag{15}$$

$$= \frac{2x}{x_0} \phi_0. \tag{16}$$

It remains to check that no additional normalization factor is needed. Using the adjoint relation,

$$\int |\hat{a}^\dagger \phi_0|^2 \, dx = \int (\hat{a}^\dagger \phi_0)^* (\hat{a}^\dagger \phi_0) \, dx \tag{17}$$

$$= \int \phi_0^* \hat{a} \hat{a}^\dagger \phi_0 \, dx. \tag{18}$$

Since $[\hat{a}, \hat{a}^\dagger] = 1$,

$$\hat{a}\hat{a}^\dagger = \hat{a}^\dagger \hat{a} + 1. \tag{19}$$

Thus

$$\int |\hat{a}^\dagger \phi_0|^2 \, dx = \int \phi_0^* \hat{a}^\dagger \hat{a} \phi_0 \, dx + \int |\phi_0|^2 \, dx \tag{20}$$

$$= \int |\hat{a} \phi_0|^2 \, dx + 1 \tag{21}$$

$$= 1. \tag{22}$$

Therefore

$$\boxed{\phi_1(x) = \hat{a}^\dagger \phi_0(x) = \frac{2x}{x_0} \phi_0(x)} \tag{23}$$

is already normalized.

(d) Raising the ground-state energy by $\hbar\omega$ gives

$$\boxed{E_1 = E_0 + \hbar\omega = \frac{3}{2}\hbar\omega.} \tag{24}$$

The Gaussian is even and the factor x is odd, so ϕ_1 is odd. It vanishes only at $x = 0$, and therefore has one node:

$$\boxed{\phi_1 : \text{ odd, with one node at } x = 0.} \tag{25}$$

Problem 2. Degeneracy and uncertainty [20 points]

(a) Acting with momentum gives

$$\hat{p}\phi_+ = -i\hbar\frac{d}{dx}e^{ikx} = +\hbar k\phi_+, \tag{26}$$

$$\hat{p}\phi_- = -i\hbar\frac{d}{dx}e^{-ikx} = -\hbar k\phi_-. \tag{27}$$

Hence the momentum eigenvalues are $+\hbar k$ and $-\hbar k$. Applying the Hamiltonian,

$$\hat{H}\phi_\pm = \frac{\hat{p}^2}{2m}\phi_\pm \tag{28}$$

$$= \frac{(\pm\hbar k)^2}{2m}\phi_\pm \tag{29}$$

$$= \frac{\hbar^2k^2}{2m}\phi_\pm. \tag{30}$$

The two linearly independent functions have the same energy, so

$$\boxed{E = \frac{\hbar^2k^2}{2m} \text{ is twofold degenerate for } k > 0.} \tag{31}$$

(b) By linearity,

$$\hat{H}\chi = \hat{H}(\phi_+ + \phi_-) \tag{32}$$

$$= E\phi_+ + E\phi_- \tag{33}$$

$$= E\chi. \tag{34}$$

Thus χ remains an energy eigenfunction. However,

$$\hat{p}\chi = \hbar k\phi_+ - \hbar k\phi_- \tag{35}$$

$$= \hbar k(\phi_+ - \phi_-). \tag{36}$$

This is not proportional to $\chi = \phi_+ + \phi_-$. Therefore

$$\boxed{\chi \text{ is an energy eigenfunction but not a momentum eigenfunction.}} \tag{37}$$

(c) Since $\hat{H} = \hat{p}^2/(2m)$,

$$[\hat{H}, \hat{p}] = \frac{1}{2m}[\hat{p}^2, \hat{p}] \tag{38}$$

$$= \frac{1}{2m}(\hat{p}^3 - \hat{p}^3) = 0. \tag{39}$$

Hence

$$\boxed{[\hat{H}, \hat{p}] = 0.} \tag{40}$$

Energy and momentum are compatible observables for the free particle. An energy measurement yielding $E = \hbar^2 k^2/(2m)$ determines only that the state lies in the eigenspace spanned by ϕ_+ and ϕ_-. It does not determine a unique vector in that space. A momentum measurement supplies the additional label:

$$\boxed{p = +\hbar k \text{ selects } \phi_+, \qquad p = -\hbar k \text{ selects } \phi_-.} \tag{41}$$

(d) Substitute $[\hat{x}, \hat{p}] = i\hbar\hat{I}$ into the general uncertainty relation:

$$\Delta x\, \Delta p \geq \frac{1}{2}\left| \int \psi^* [\hat{x}, \hat{p}] \psi \, dx \right| \tag{42}$$

$$= \frac{1}{2}\left| i\hbar \int |\psi|^2 \, dx \right|. \tag{43}$$

Since the state is normalized,

$$\int |\psi|^2 \, dx = 1. \tag{44}$$

Therefore

$$\boxed{\Delta x\, \Delta p \geq \frac{\hbar}{2}.} \tag{45}$$

The lower bound follows from the nonzero commutator, not from imperfect measuring instruments.

Problem 3. Attractive delta-well bound state [20 points]

(a) A bound-state wave function must decay as $|x| \to \infty$. Since the outside potential is zero, this requires

$$\boxed{E < V_{\text{outside}} = 0.} \tag{46}$$

For $x \neq 0$, the delta potential vanishes and the Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\phi'' = E\phi. \tag{47}$$

With $\alpha = \sqrt{-2mE}/\hbar$, this is

$$\phi'' = \alpha^2 \phi. \tag{48}$$

The general exponential solutions are restricted by normalizability. On the left, only $e^{\alpha x}$ decays as $x \to -\infty$; on the right, only $e^{-\alpha x}$ decays as $x \to +\infty$. Thus

$$\phi(x) = \begin{cases} Ce^{\alpha x}, & x < 0, \\ De^{-\alpha x}, & x > 0. \end{cases} \tag{49}$$

Continuity at $x = 0$ gives $C = D = A$, and hence

$$\boxed{\phi(x) = Ae^{-\alpha|x|}.} \tag{50}$$

(b) Integrating the Schrödinger equation through the origin gives

$$-\frac{\hbar^2}{2m}\int_{-\varepsilon}^{\varepsilon} \phi''(x)\,dx - g\int_{-\varepsilon}^{\varepsilon} \delta(x)\phi(x)\,dx = E\int_{-\varepsilon}^{\varepsilon} \phi(x)\,dx. \tag{51}$$

Evaluate the first two integrals:

$$-\frac{\hbar^2}{2m}[\phi'(\varepsilon) - \phi'(-\varepsilon)] - g\phi(0) = E\int_{-\varepsilon}^{\varepsilon} \phi(x)\,dx. \tag{52}$$

For a finite continuous wave function, the integral on the right approaches zero as $\varepsilon \to 0$. Therefore

$$-\frac{\hbar^2}{2m}[\phi'(0^+) - \phi'(0^-)] - g\phi(0) = 0. \tag{53}$$

Solving for the derivative jump gives

$$\boxed{\phi'(0^+) - \phi'(0^-) = -\frac{2mg}{\hbar^2}\phi(0).} \tag{54}$$

(c) For $\phi(x) = Ae^{-\alpha|x|}$,

$$\phi'(0^+) = -\alpha A, \qquad \phi'(0^-) = +\alpha A, \qquad \phi(0) = A. \tag{55}$$

The jump condition becomes

$$-2\alpha A = -\frac{2mg}{\hbar^2}A. \tag{56}$$

Since a physical state has $A \neq 0$,

$$\boxed{\alpha = \frac{mg}{\hbar^2}.} \tag{57}$$

From $E = -\hbar^2\alpha^2/(2m)$,

$$\boxed{E = -\frac{mg^2}{2\hbar^2}.} \tag{58}$$

(d) Normalize the wave function:

$$1 = \int_{-\infty}^{\infty} |\phi(x)|^2\, dx \tag{59}$$

$$= 2|A|^2 \int_0^{\infty} e^{-2\alpha x}\, dx \tag{60}$$

$$= 2|A|^2 \frac{1}{2\alpha} \tag{61}$$

$$= \frac{|A|^2}{\alpha}. \tag{62}$$

Choosing A real and positive gives

$$A = \sqrt{\alpha}. \tag{63}$$

Therefore the normalized bound state is

$$\boxed{\phi(x) = \sqrt{\alpha}\, e^{-\alpha|x|}, \qquad \alpha = \frac{mg}{\hbar^2}.} \tag{64}$$

The jump condition selects one and only one positive value of α, and therefore one bound-state energy. Thus

$$\boxed{\text{There is exactly one bound state and no excited bound states.}} \tag{65}$$